



Cairo University - Faculty Of Engineering
Computer Engineering Department
Graduation Project - Fall 2025



بَلِغ

Document of Deliverables

Baligh: Arabic Writing Assistant

Supervised by

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December 23, 2025

1 Problem Statement

While English speakers enjoy seamless writing assistants, Arabic users lack tools that have Grammatical Error Correction and Next-Word Suggestion with high accuracy and without significant input latency.

2 Motivation

We aim to build a high-quality MSA writing assistant that achieves SOTA accuracy with real-time performance, bridging the gap between current research advancements and available practical tools.

3 System Architecture

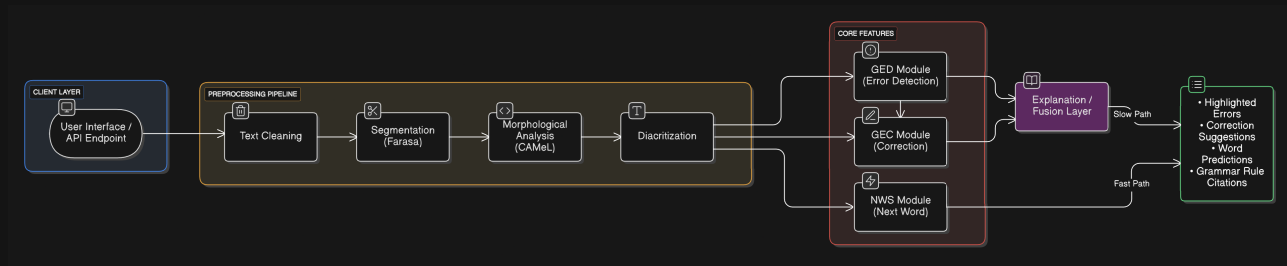


Figure 1: High-Level System Architecture of Baligh

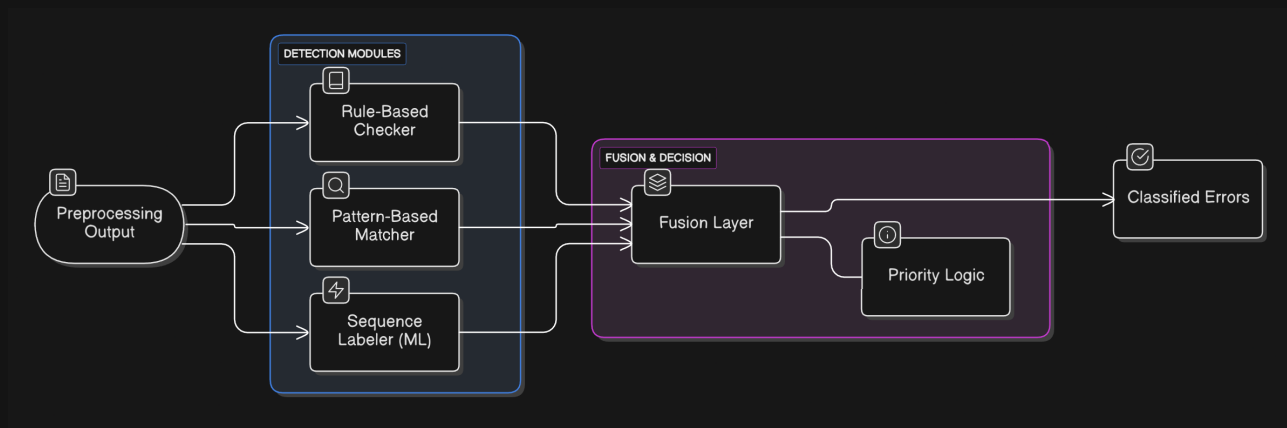


Figure 2: Grammatical Error Detection Architecture

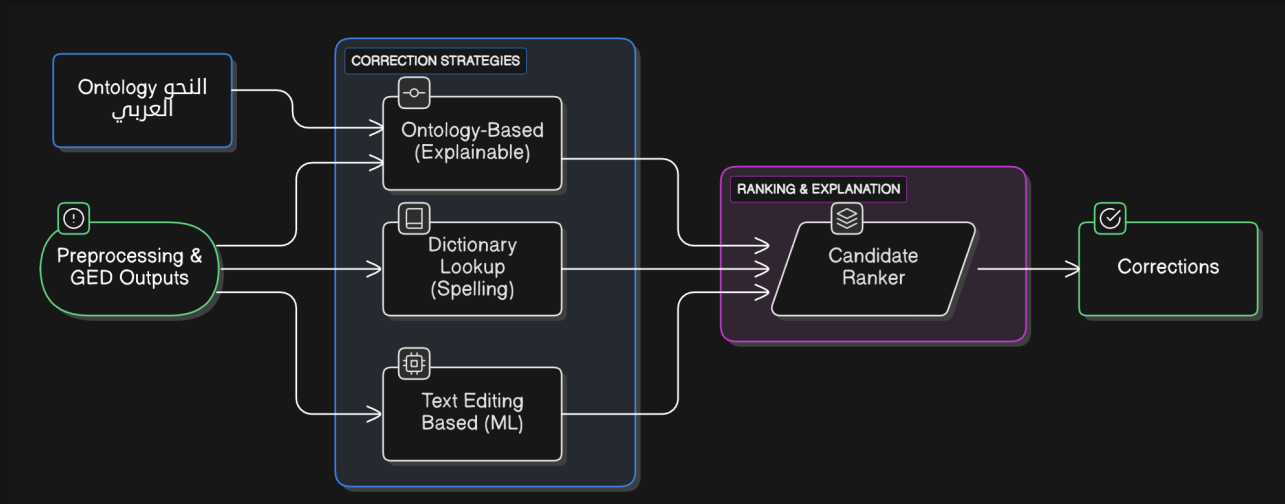


Figure 3: Grammatical Error Correction Architecture

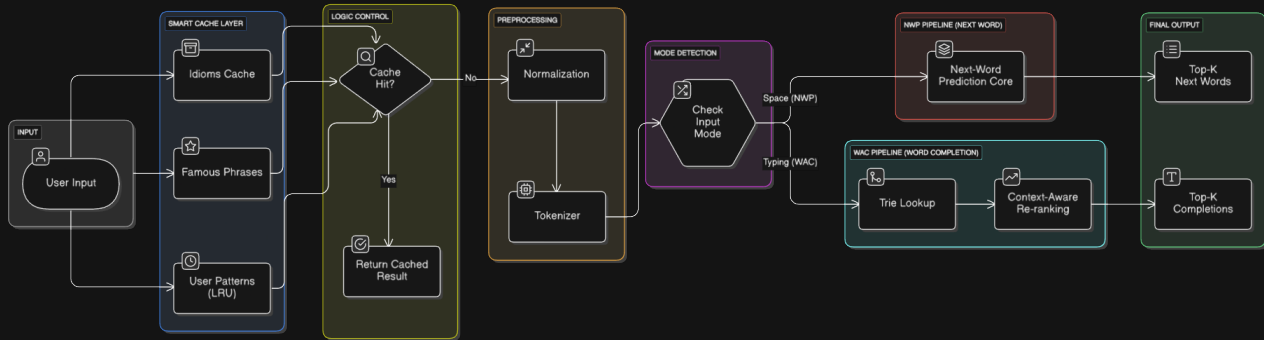


Figure 4: Next Word Suggestion Architecture

4 List of Deliverables

Module Name	Function	Input	Expected Output	% Libs
1. Preprocessing Module	Text cleaning, normalization, tokenization, and morphological analysis	User Input (Raw Arabic Text)	Tokenized text + Morphological Features (POS, SP, Lemma)	10%
2. GED Engine	Error Detection (Rules + Patterns + ML + Fusion)	Preprocessed Input	Final error list with confidence & explanations	20%
3. GEC (Ontology & Dictionary-Based)	Generate corrections using grammar ontology and spelling mistakes using dictionary	Preprocessed Input	Correction candidates	20%
4. GEC (Text Editing & Ranking)	Text editing approach + candidate ranking + explanation generation	Preprocessed Input + M4 Correction Candidates	Final ranked corrections with explanations	15%
5. NWS Module	Next Word Suggestion	Preprocessed Input	Ranked list of predicted words	30%
6. API & GUI	REST API + Frontend	User Inputs + All Modules Outputs	Web UI with a live text editor + (Optional Browser Extension)	20%